

Practice Quiz 06B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 06

Existence and uniqueness as two separate claims.

Label and prove both halves.

1. Prove that for every real number $a \neq 0$, there is exactly one real number x such that $3ax - 2 = 7$.

Solution Let $a \in \mathbf{R}$ and suppose $a \neq 0$.

Existence. Set $x = 3/a$. This is real because $a \neq 0$, and

$$3ax - 2 = 3a \left(\frac{3}{a} \right) - 2 = 9 - 2 = 7.$$

Thus a solution exists.

Uniqueness. Suppose x and y both satisfy the equation. Then $3ax - 2 = 7 = 3ay - 2$, so $3a(x - y) = 0$. Since $3a \neq 0$, it follows that $x = y$. Thus the solution is unique.

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